Software Requirements Specification

for

Play, Learn, and Protect Platform

Version 1.0 approved

<12/15/2025>

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1. Introduction

1.1 Purpose

The purpose of this document is to present a detailed description of the Play, Learn, and Protect Platform. It will explain the purpose and features of the system, the interfaces of the system, what the system will do, operational parameters, and constraints, acting as a reference for developers, testers, project managers, and involved parties during development.

1.2 Document Conventions, Definitions, and Abbreviations

Functional Requirements	Prefixed with FR (e.g., FR-01)
Non-Functional Requirements	Prefixed with NFR (e.g., NFR-01)
Specifications	Prefixed with SP (e.g., SP-01)
User Stories	Prefixed with US (e.g., US-01)
Essential terminology and section titles	Appear in **Bold**
Importance tiers/ Prioritization	Categorized as Must, Should, or Could
PLP	Play, Learn, and Protect
REST	Representational State Transfer, a way to structure APIs
API	Application Programming Interface
HTTPS	Hyper Test Transfer Protocol Secure
JSON	JavaScript Object Notation

1.3 Product Scope

The software being produced is called the Play, Learn, and Protect system. The system will allow creativity and social skills through playful games that prioritizes Egyptian culture, and ensure digital safety. This creates a cultural first, engaging, and protected online environment for Egyptian youth aged 3–12.

The PLP system will allow a user to register as a parent or a teacher. The parent will be allowed to create a child account, and set the privacy and permission options. A parent will have the option to freeze a child's account, and overview screen-time, digital behavior, and performance through their dashboard. The platform will allow children users to open the dashboard and select a game from a wide variety of age-appropriate games, a child will only be able to select or access games that are within the permissions set by their parent and appropriate for their age. The platform will allow children users to receive points according to their accomplishment and unlock achievement badges, and view where their rank amongst their peers (if allowed by the parent). The platform will allow teachers to apply for a teachers account, provided that they work for an educational institution, which will be confirmed by the admin before account creation. The platform will allow teachers to recommend games or modules for children under their supervision, all children under a teachers' supervision must be confirmed by their parents. Teachers will also be allowed to overview screen-time, digital behavior, performance of students that are under their supervision. The admin will be allowed to review and accept/reject teacher proposals. The admin can ban any user, specifying a reason. The admin can add/remove games/modules.

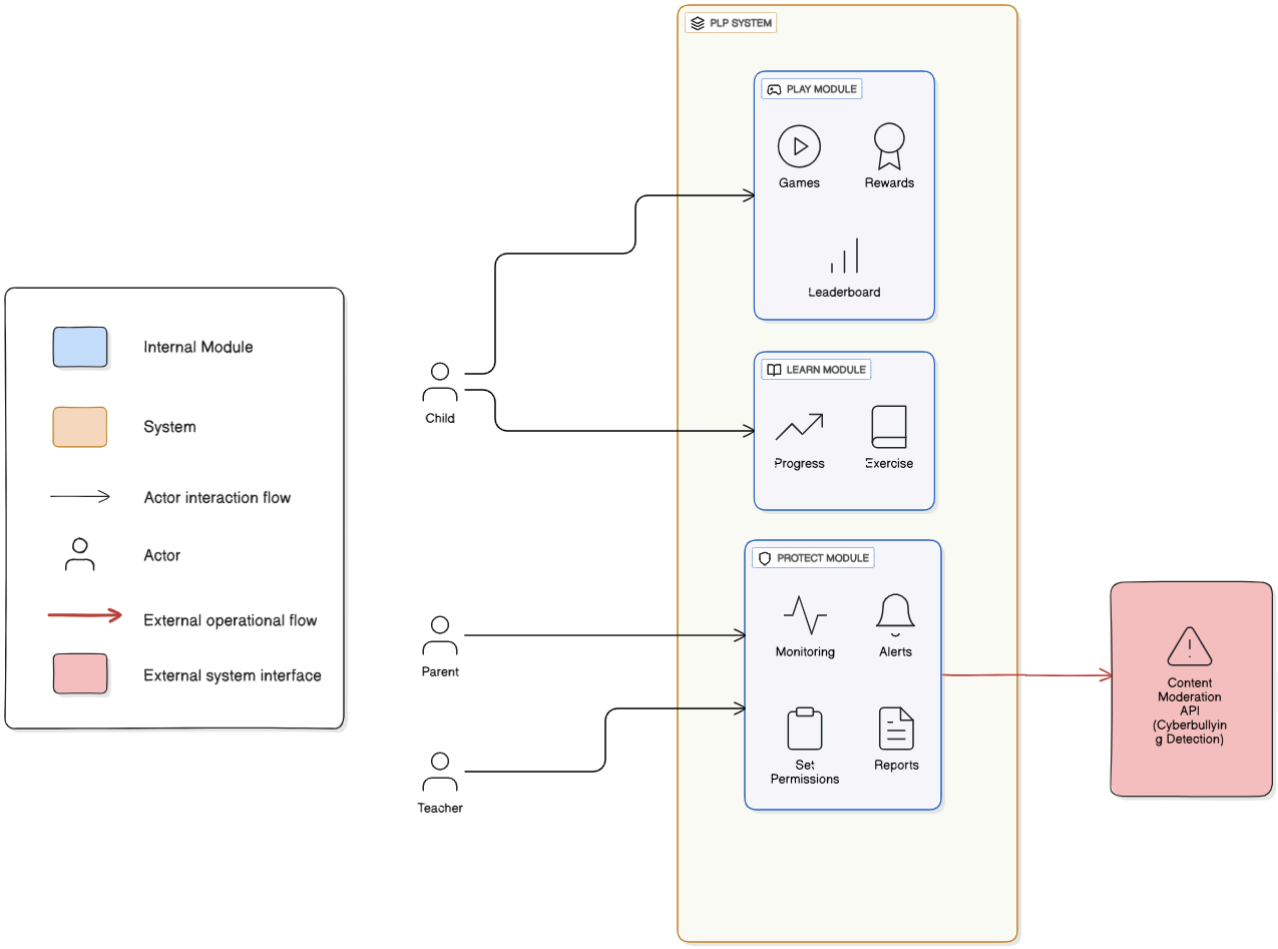
The system has multiple constraints on what it can provide. The system will not provide educational material in the form of videos. It will not allow parents or teachers to overview children that are not under their supervision. The platform will not allow children to create accounts, or take actions that exceed their allowed permissions set by their parents. Teachers will not be allowed to recommend content to students that isn't appropriate to their age. There will only be one admin that is hard-coded by the system.

1.4 References

IEEE. *IEEE Std 830-1998 IEEE Recommended Practice for Software Requirements Specifications*. IEEE Computer Society, 1998.

2. Overall Description

2.1 Product Perspective



In the diagram above the fully integrated PLP system is illustrated. The system has 3 active actors, each interacting with their respective modules (shown via black arrows). Interactions with external systems are shown via red arrows. External systems are placed inside a red box. All internal modules are placed inside the orange box (labelled as PLP system). Internal modules are placed in blue boxes within the PLP system.

2.2 Product Functions

FR-ID	Description	Priority
FR-1	Allow children to select and launch age-appropriate educational games.	Must
FR-2	Track and reward points for completing games.	Must
FR-3	Display a leaderboard to show top learners per age group.	Should
FR-4	Apply learned knowledge through a gamified experience in.	Must
FR-5	Allow parents/teachers to monitor a child's screen time	Must
FR-6	Allow parents/teachers to monitor a child's digital behavior	Must
FR-7	Allow parents/teachers to monitor a child's performance.	Must
FR-8	Alert parents/teachers of potential cyberbullying	Must
FR-9	Allow system administrators to upload and manage educational content.	Must
FR-10	Allow multi-role login (Child, Parent, Teacher, Admin).	Must
FR-11	Allow children to save and resume progress in learning modules.	Must
FR-12	Let parents set daily screen time limits per child account.	Must
FR-13	Provide help within games when a child is stuck.	Could

FR-14	Allow parental accounts to manage multiple child profiles.	Must
FR-15	Include culturally relevant themes games.	Must
FR-16	Notify parents when a safety alert is triggered.	Must

2.3 User Classes and Characteristics

There are 4 types of users that will interact with our platform. Child users (aged 3-12) are assumed to have mixed varying digital literacy and short attention spans. Parent users are assumed to have potentially restricted technical skills, are safety-conscious, and want visibility into a child's activity, and performance. Teacher users are curriculum experts, they oversee multiple students, and overview their reports. System administrators are responsible for account governance, content curation, and infrastructure upkeep.

2.4 Design Constraints

There are constraints to be considered when developers are developing the platform. All content must be delivered primarily in Arabic and reflect Egyptian culture. Easy to use interface for children such as large touch targets and intuitive navigation. Parental consent is required for data collection on their children.

2.5 Assumptions and Dependencies

The system assumes children will have access to an internet-connected device, and parents and teachers have basic digital literacy. It depends on stable cloud infrastructure for hosting, integration with third-party content moderation APIs for safety features, and ongoing educator collaboration to validate curriculum alignment and cultural relevance.

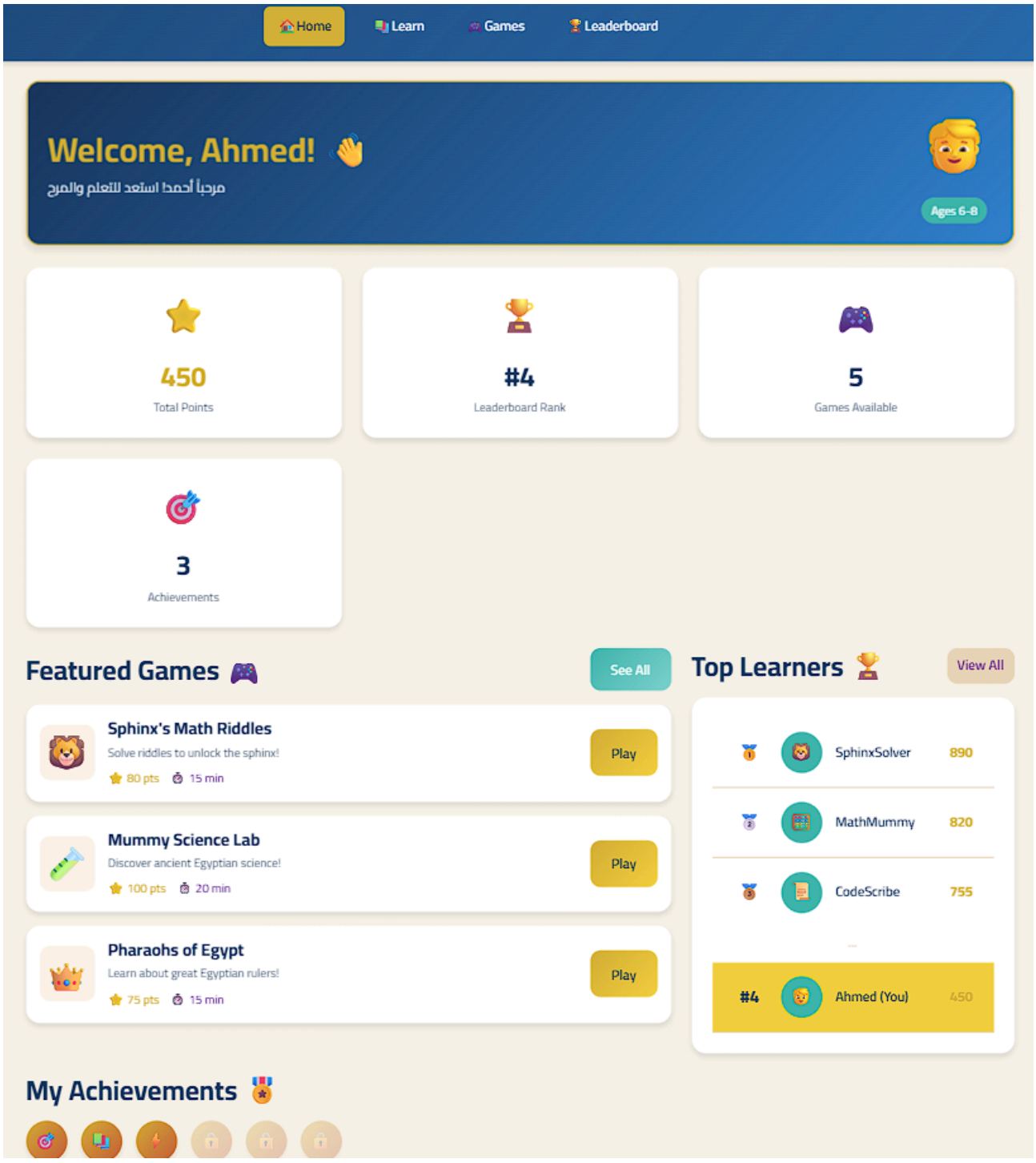
3. External Interface Requirements

3.1 User Interfaces

The Play, Learn & Protect platform features four unique, role-specific interfaces, each crafted with well-defined logical traits, accessibility criteria, and cultural adaptation for users in Egypt.

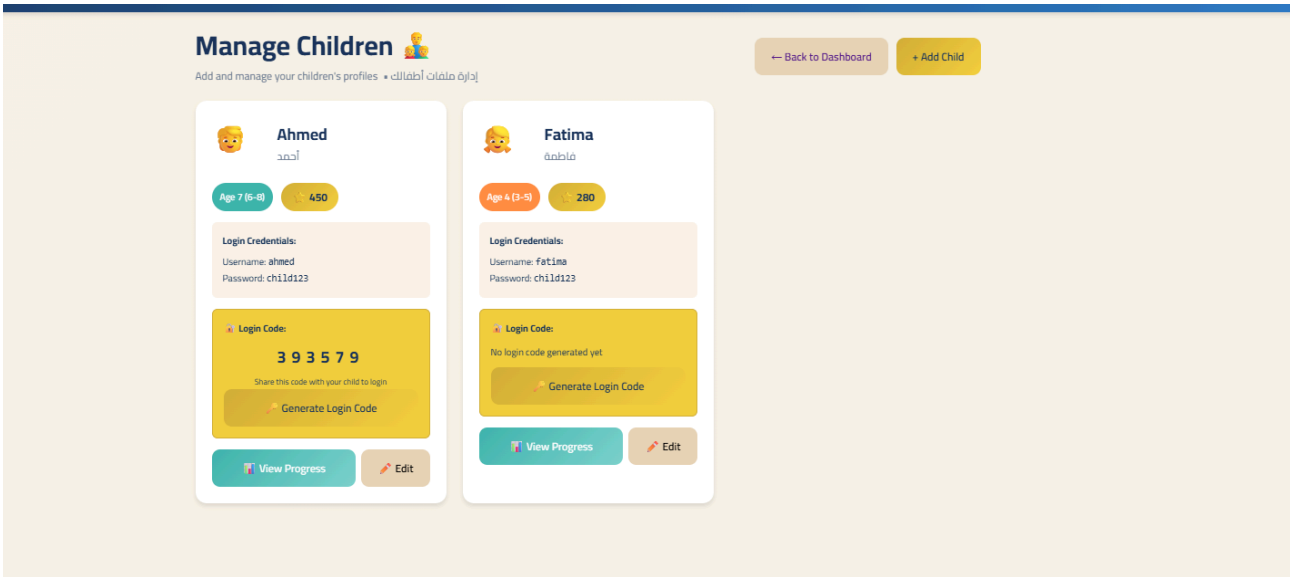
3.1.1) **Child's Interface**

The children interface is designed to be minimal and intuitive, it includes minimal navigation and large interactive elements. Its visual theme incorporates bright, culturally relevant Egyptian characters and environments.



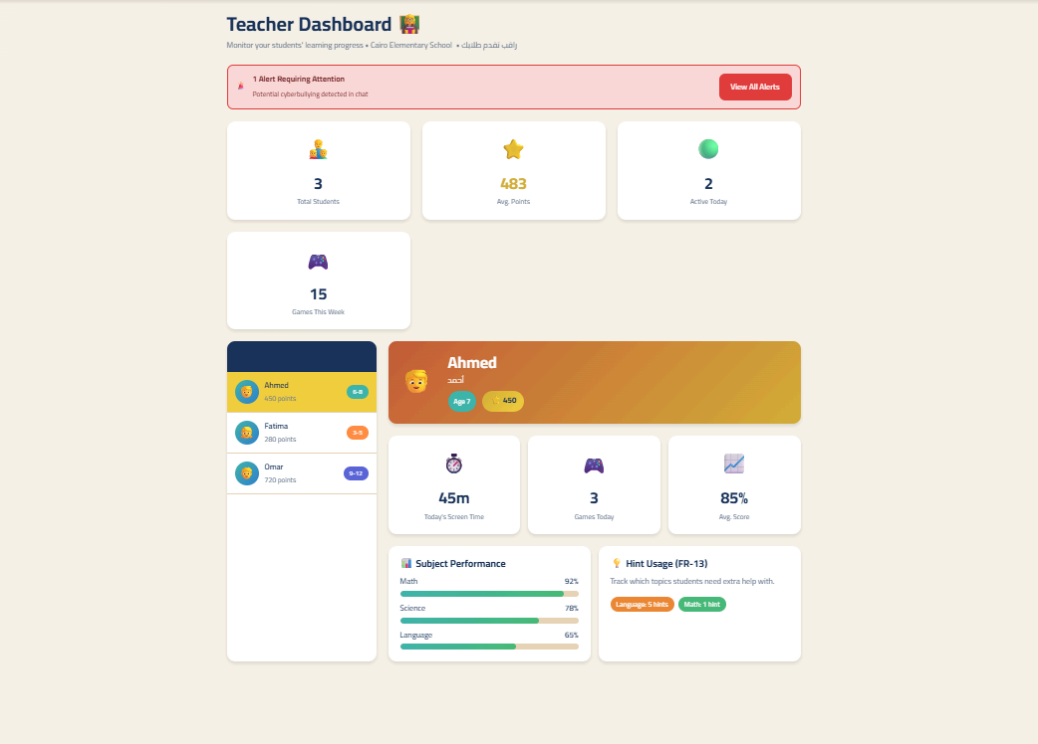
3.1.2) **Parents Interface**

The parent dashboard features a flexible, widget-oriented design that supports personalization via drag-and-drop capability. It embraces a sleek, professional look featuring integrated data visualizations like time-series graphs and progress charts. Navigation is facilitated by a uniform top bar featuring links to Dashboard, Children, Reports, Settings, and Help.



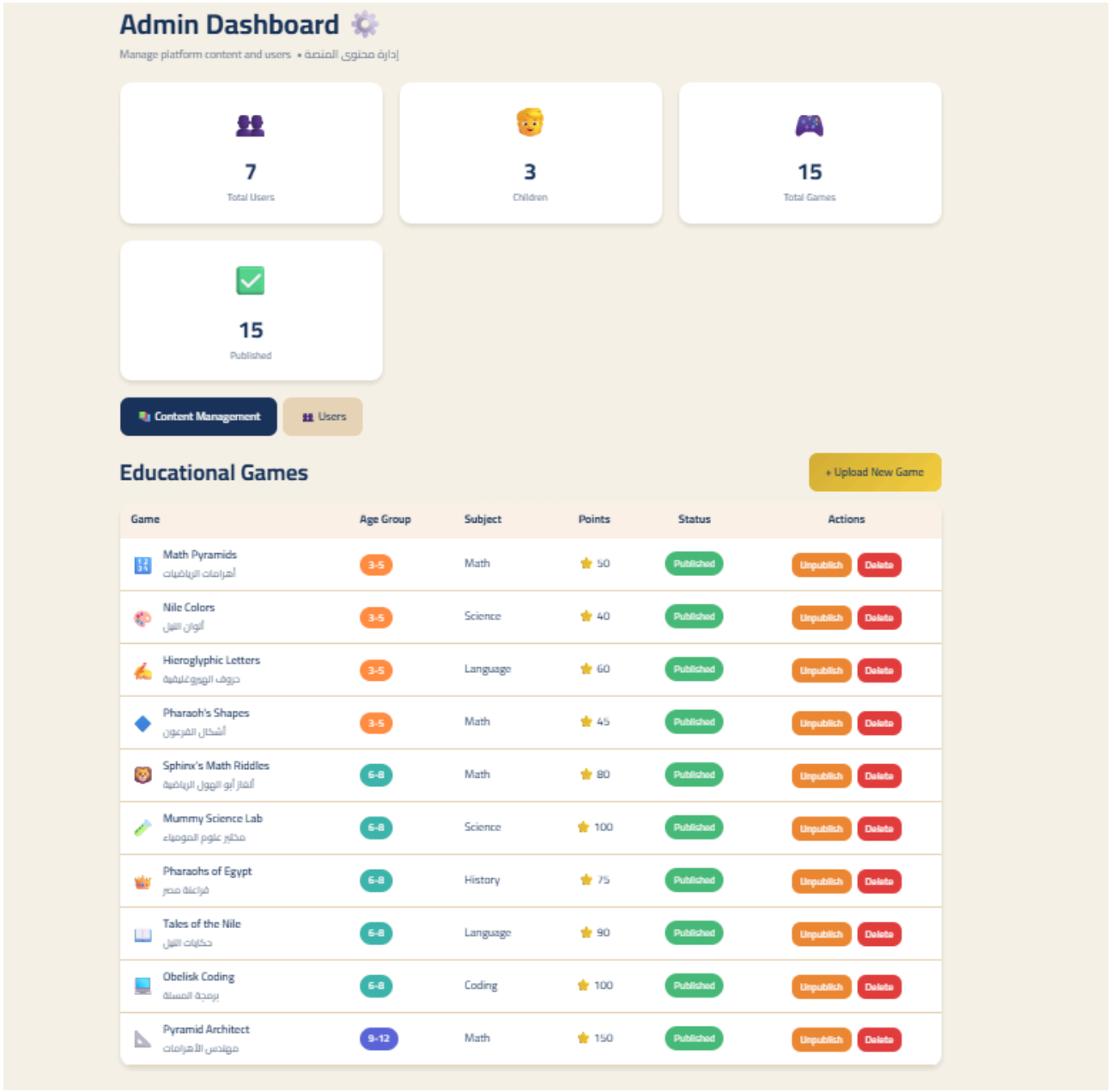
3.1.3) **Teacher Dashboard**

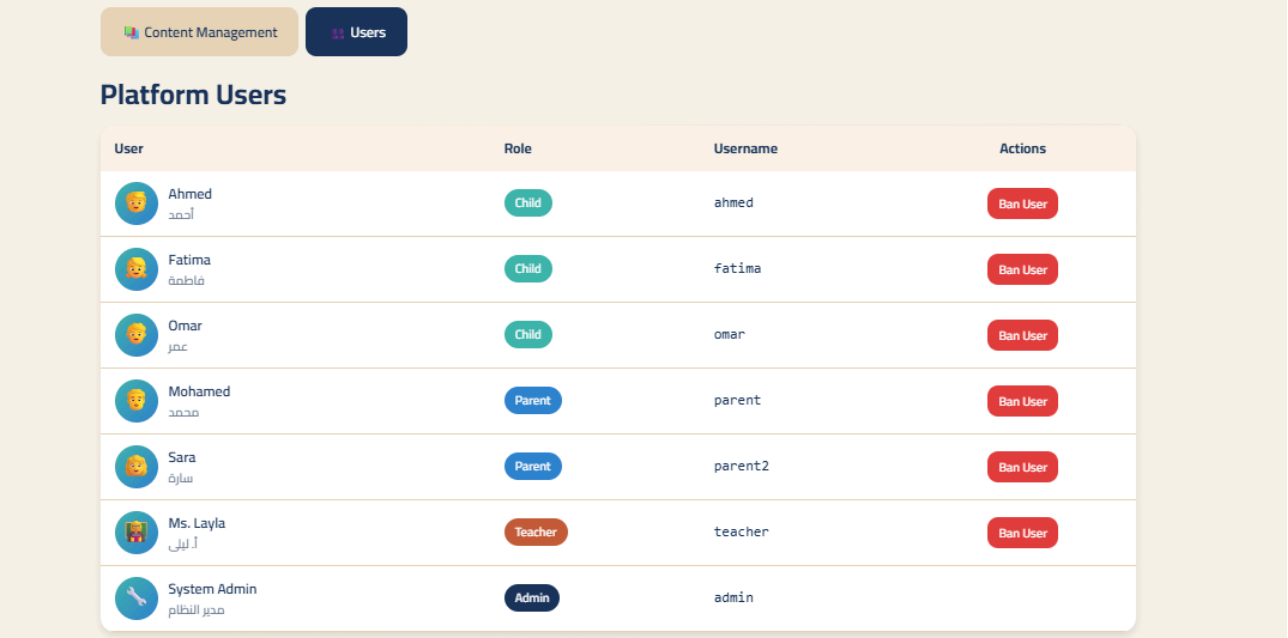
The teacher interface features a multi-panel design optimized for classroom management, with simultaneous views of the class roster, individual student cards, and aggregate analytics.



3.1.4) **Admin Dashboard**

The administrator interface is a multi-tab control panel built for system oversight, user management, and content moderation. It employs a technical but clear layout, with advanced filtering, search capabilities, and a logical action hierarchy. Primary navigation is organized into top-level tabs: Users, Content, Analytics, System, and Audit Logs.


Admin Dashboard
Manage platform content and users • إدارة محتوى المنصة
7
Total Users
3
Children
15
Total Games
15
Published
Content Management
Users
Educational Games
+ Upload New Game
Game
Age Group
Subject
Points
Status
Actions
Math Pyramids
أهرامات الرياضيات
3-5
Math
50
Published
Unpublish
Delete
Nile Colors
ألوان النيل
3-5
Science
40
Published
Unpublish
Delete
Hieroglyphic Letters
حروف الهيروغليفية
3-5
Language
60
Published
Unpublish
Delete
Pharaoh's Shapes
أشكال الفرعون
3-5
Math
45
Published
Unpublish
Delete
Sphinx's Math Riddles
ألغاز أبو الهول الرياضية
6-8
Math
80
Published
Unpublish
Delete
Mummy Science Lab
مختبر علوم المومياء
6-8
Science
100
Published
Unpublish
Delete
Pharaohs of Egypt
فراعنة مصر
6-8
History
75
Published
Unpublish
Delete
Tales of the Nile
حكايات النيل
6-8
Language
90
Published
Unpublish
Delete
Obelisk Coding
برمجة المسلة
6-8
Coding
100
Published
Unpublish
Delete
Pyramid Architect
مهندس الأهرامات
9-12
Math
150
Published
Unpublish
Delete



3.2 Software Interfaces

Supabase, a backend-as-a-service (BaaS), will be primarily used as the backend. It offers multiple services that will help speed the development of the project:

- A REST API (via PostgREST): used to communicate with the PostgreSQL database.
- Auth: Supabase uses JWT as its authentication strategy, and already provides a read-to-use register and login functionality that follows security best practices.
- PostgreSQL database: Supabase automatically handles the deployment of the database.
- File storage: Supabase has a file storage implementation, which eliminates having to use an external file storage service.
- Realtime support: updates to the database are immediately sent to subscribed clients.

All communication is done via HTTPS (except for Supabase Realtime, it uses Websockets), and request/response bodies are in JSON format.

We will use two methods for content moderation:

1. Google's Perspective API: used to detect cyberbullying and inappropriate content
    - Six detection categories of toxicity: (toxicity, severe toxicity, identity attack, insult, threat, profanity)
    - Provides multilingual support (for Arabic).
2. Our own custom-built time tracker to detect excessive gaming.

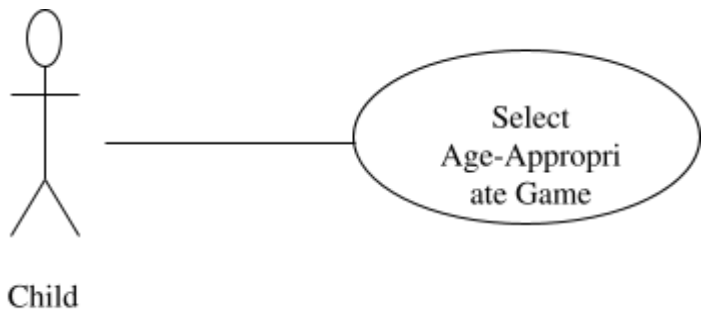
4. Functional Requirements

4.1) **FR-1**: Allow children to select and launch age-appropriate educational games (3–12 years).
User Story
As a child user, I want to be able to choose and start games that are right for my age group, so that I can learn and play safely without seeing content that's too easy or too hard for me.

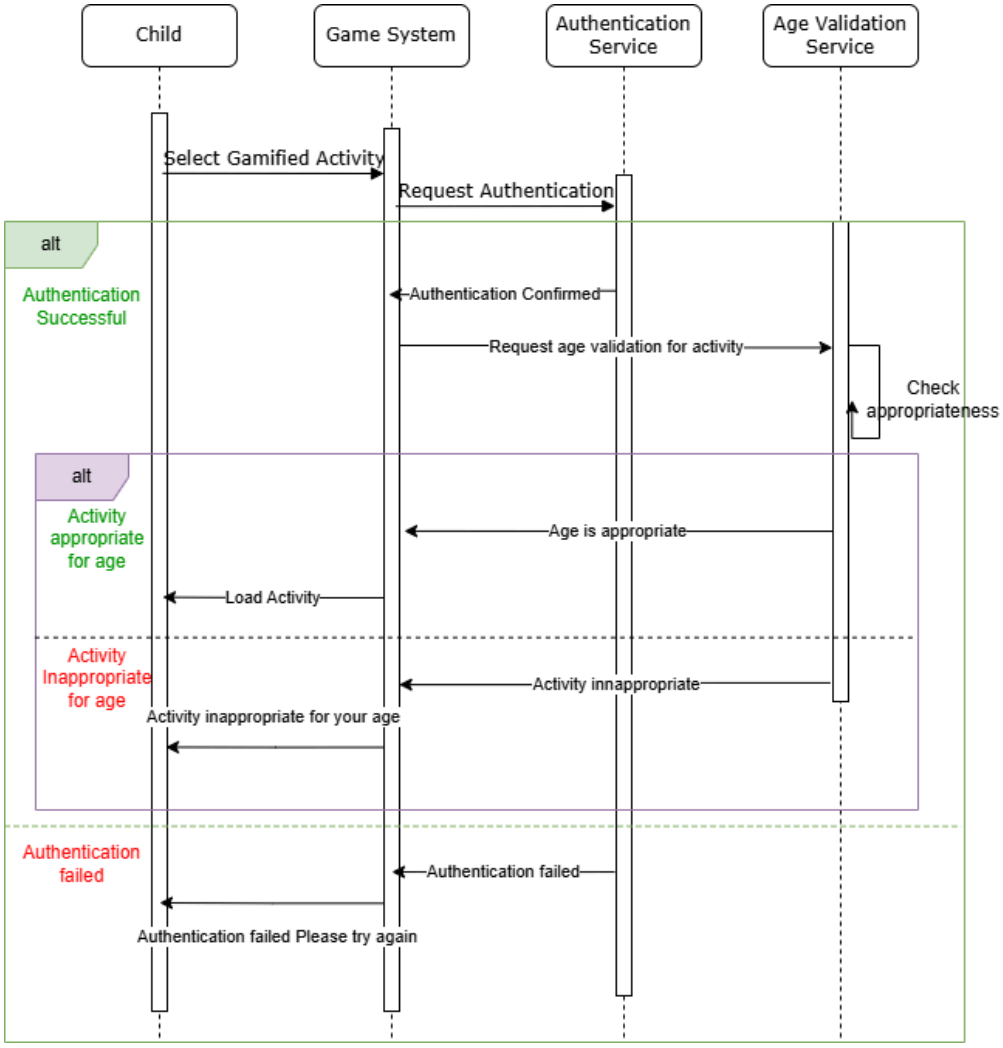
4.1.1) **Specifications**

ID	Specification
SP-1-01	The system shall authenticate the child before allowing access to the game library.
SP-1-02	The child's age group shall be retrieved from their profile upon login.
SP-1-03	The game library shall be filtered dynamically based on the child's age group.
SP-1-04	Each game tile shall display: title, age badge, learning objective, and estimated playtime.
SP-1-05	Upon game selection, the system shall validate that the game's age range matches the child's age group.
SP-1-06	Game launch shall initialize within 3 seconds after selection.
SP-1-07	The child shall be able to exit the game and return to the library at any time.
SP-1-08	All game launch attempts shall be logged for parental review.

4.1.2) **UC-01**



4.1.3) **Sequence Diagram**

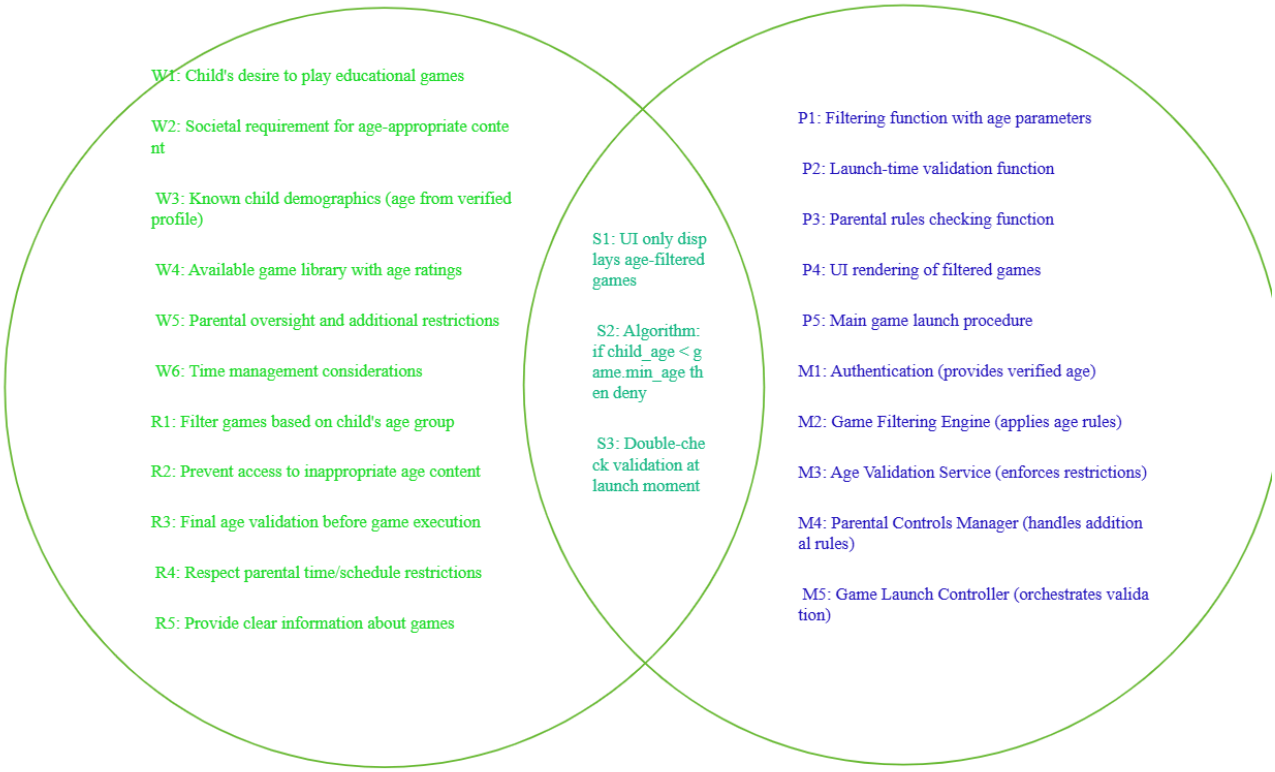


4.1.4) **Acceptance Criteria**

Acceptance Criterion	**Test Scenario**
Given the child is logged in, when they view the game library, then only games for their age group are displayed.	Login as 6-year-old → see only 6–8 games.
Given a child selects a game, when the game's age range does not match their profile, then an age-restriction message is shown and the game does not launch.	5-year-old tries to launch a 9–12 game → error message appears.
Given a valid game selection, when the child clicks "Play", then the game loads within 3 seconds.	Timer from click to game start screen ≤ 3 seconds.

Given the game is playing, when the child clicks “Exit”, then they are returned to the game library.	Exit button functions and returns to the previous view.
Given any game launch attempt, when completed, then an entry is recorded in the activity log with correct details.	The parent dashboard shows game played, time, duration.
Given the child is not authenticated, when they try to access the game library, then they are redirected to the login screen.	Direct URL access without login → redirect to login.
Given the child’s profile age is updated (e.g., birthday), when they next login, then the game library reflects the new age group.	Age updated from 8 to 9 → library now shows 9–12 games.

4.1.5) **WRSPM Diagram**



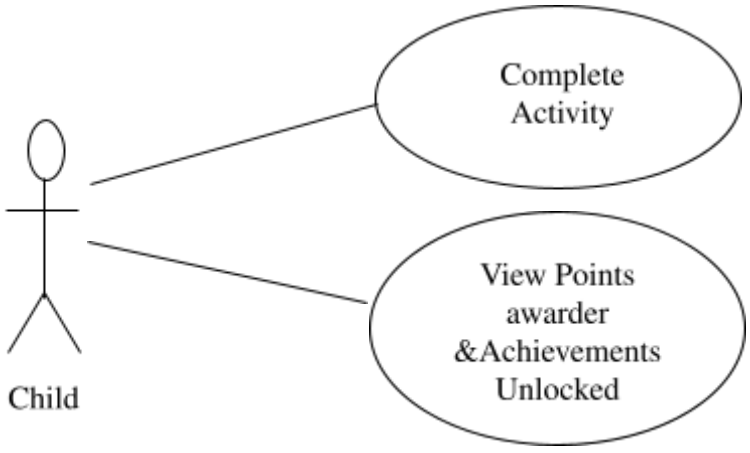
eh: Child's actual developmental level, parent's judgment of appropriate content
ev: Child's age from profile, selected game ID, library display
sv: Game library interface, age badges, launch buttons, exit option
sh: Age validation logic, filtering algorithms, logging mechanism

4.2) **FR-2**: Track and reward points for completing gamified activities.
User Story
As a child user, I want to earn points when I finish learning activities and mini-games, so that I feel motivated to keep learning and can see my progress grow over time.

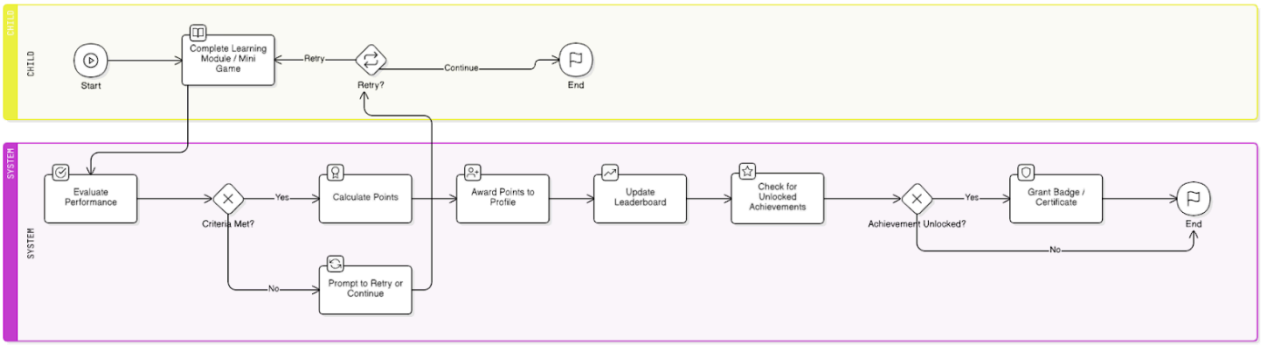
4.2.1) **Specifications**

ID	Specification
SP-2.1	The system shall track completion of each learning module and mini-game.
SP-2.2	Points shall be awarded based on: · Answer Accuracy · Completion Time · Module Difficulty
SP-2.3	The point breakdown shall be displayed to the child after module completion.
SP-2.4	Points shall be added in the child's profile and visible in their dashboard.
SP-2.5	Achievements/badges shall be unlocked when point milestones are reached.
SP-2.6	Parents/teachers shall receive notifications for major achievements.

4.2.2) **UC-02**



4.2.3) **BPMN**



4.2.4) **Acceptance Criteria**

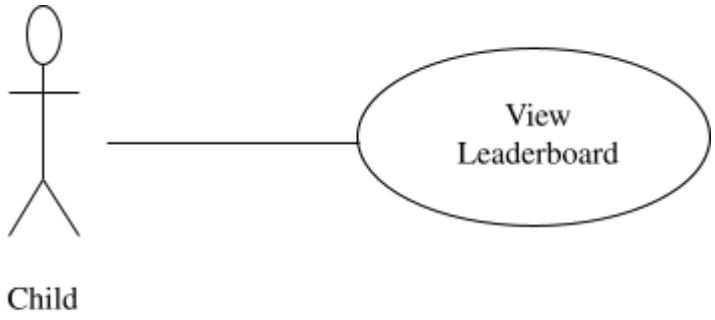
Acceptance Criterion	**Test Scenario**
Given a child completes a learning module, when they finish, then points are calculated and displayed within 5 seconds.	Complete a math game -> see points breakdown popup.
Given points are awarded, when the child views their profile, then the total points are updated correctly.	Check profile before/after earning 50 points -> total increases by 50.
Given a child earns 100 total points, when the milestone is reached, then the "Century Club" badge is unlocked.	Points go from 99 to 100 -> badge appears in achievements.
Given a child is in the 6–8 age group, when they earn points, then their rank updates on the age-group leaderboard.	Earn points -> check leaderboard -> see improved rank.
Given a parent has notifications enabled, when their child unlocks a major achievement, then the parent receives an alert.	Child earns "Science Star" -> parent gets in-app notification.
Given the system experiences a failure during point awarding, when recovered, points are not duplicated or lost.	Simulate crash after module completion -> ensure points awarded once.

4.3) **FR3**: Leaderboard Functionality
User Story
As a child user, I want to see a leaderboard that shows the top learners in my age group, so that I can feel motivated and compare my progress with others safely.

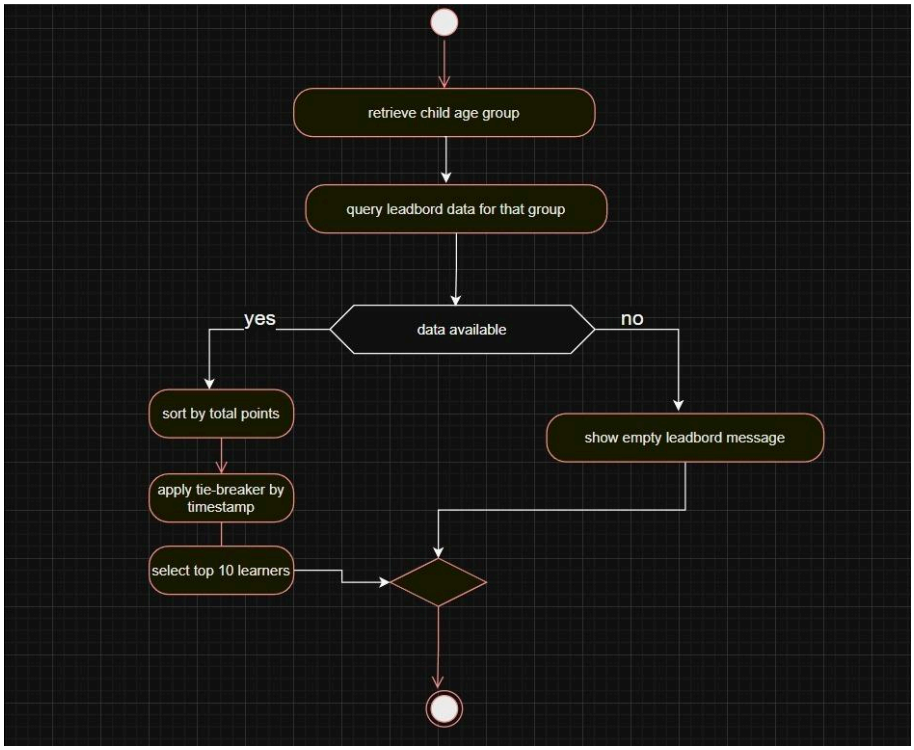
4.3.1) **Specifications**

ID	Specification	Notes
SP-3.1	The system must create leaderboards for every age category (3–5, 6–8, 9–12).	Sorted according to profile age.
SP-3.2	Rankings must be determined by the accumulated points.	Points earned from learning modules and mini-games.
SP-3.3	The leaderboard must update no less than once every day.	Automated background task.
SP-3.4	Display solely the leading 10 learners within each age category.	Maintains confidentiality and clarity.
SP-3.5	Every record must show the child's nickname, overall points, and badge rank.	Actual names concealed for confidentiality.
SP-3.6	Only designated roles should have access to view entries along with student IDs.	Access limited to roles (e.g., Teachers).
SP-3.7	If two participants obtain equivalent scores, the ranking preference is assigned to the one who achieved it first.	Rule for tie-breaking.

4.3.2) UC-03



4.3.3) **Diagram**



4.3.4) **Acceptance Criteria**

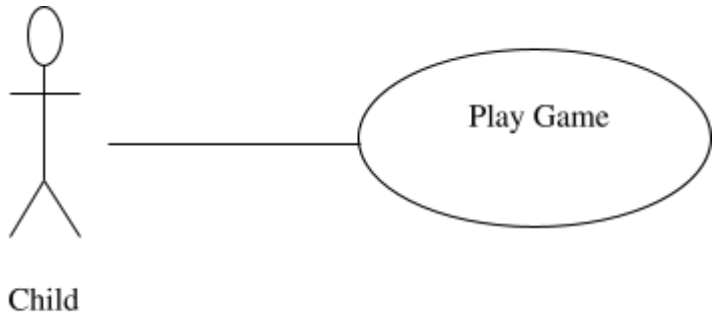
Acceptance Criterion	**Test Scenario**
Leaderboard displays the highest achievers for the appropriate age bracket.	Log in as a 6-year-old → view leaderboard for 6–8 age category only.
Positions are determined by points.	Verify user scores → confirm rankings align.
Leaderboard displays at most 10 entries.	Add >10 dummy accounts → verify only 10 shows.
When ties occur, priority goes to the one who achieved the score earlier.	Two users with equal points → verify their sequence based on timestamp.
Teachers and parents have the ability to see the leaderboard with student ID tags.	Log in as a teacher → confirm that IDs are displayed.
Real names are not shown to child users.	Log in as child → only nicknames displayed.

4.4) **FR4**: Gamified Learning Activities.
User Story
As a child user, I want to apply what I learned through fun gamified activities, so that I can practice my knowledge in an enjoyable and interactive way.

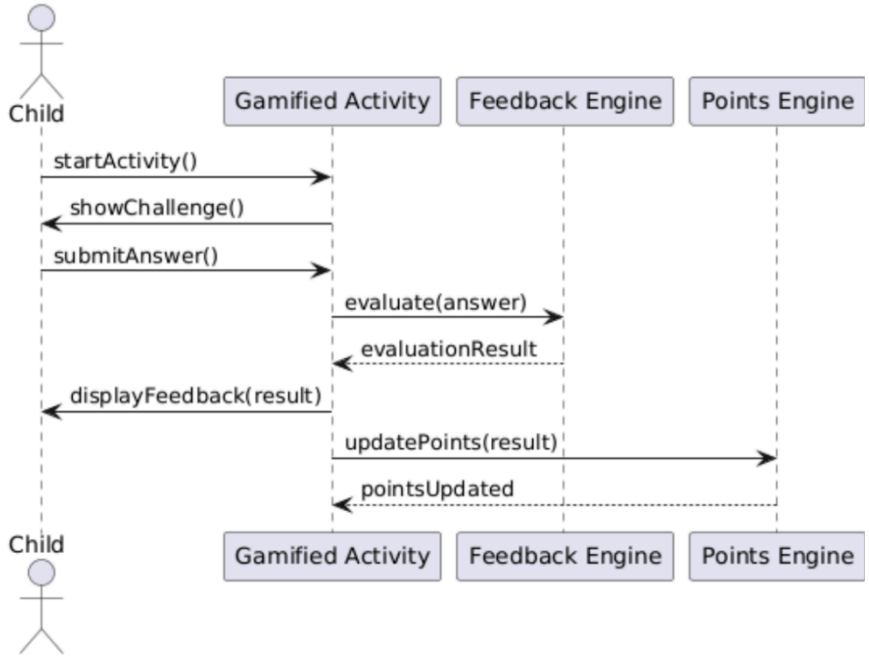
4.4.1) **Specifications**

ID	Specification	Notes
SP-4.1	The system must offer game challenges corresponding to subjects (math, science, language, coding).	Learning goals linked to tasks.
SP-4.2	Tasks ought to modify their complexity according to the child's performance.	Adaptive learning that is dynamic.
SP-4.3	Every gaming interaction must offer feedback (right or wrong).	Visual/Audio signals for feedback.
SP-4.4	The system must monitor child responses for learning analytics.	Saved for teacher analysis.
SP-4.5	Completing activities should add to the child's points.	Teachers have access to notifications.
SP-4.6	Visuals and interactive elements in game tasks must be appropriate for children aged 3–12.	User-friendliness and accessibility focus.
SP-4.7	Tasks should incorporate milestones to safeguard against losing progress.	Assists in monitoring performance and preventing lost efforts.

4.4.2) **UC-04**



4.4.3) **Sequence Diagram**



Acceptance Criterion	**Test Scenario**
Tasks correspond to educational subjects.	Open math module → check that math-related activities are present.
Feedback is provided after each reply.	Submit a response → feedback shows up instantly.
The challenge level modifies according to performance.	Maintain a winning streak → questions become more difficult.
Completing the activity raises the score.	Complete activity → total score rises.

Progress markers operate as intended.	Leave activity → return → resume from marker.
The visual design suits the age group.	UX assessment focused on ages 3–12 confirms friendly interface.

4.5) **FR5:** Allow parents/teachers to monitor a child's screen time.

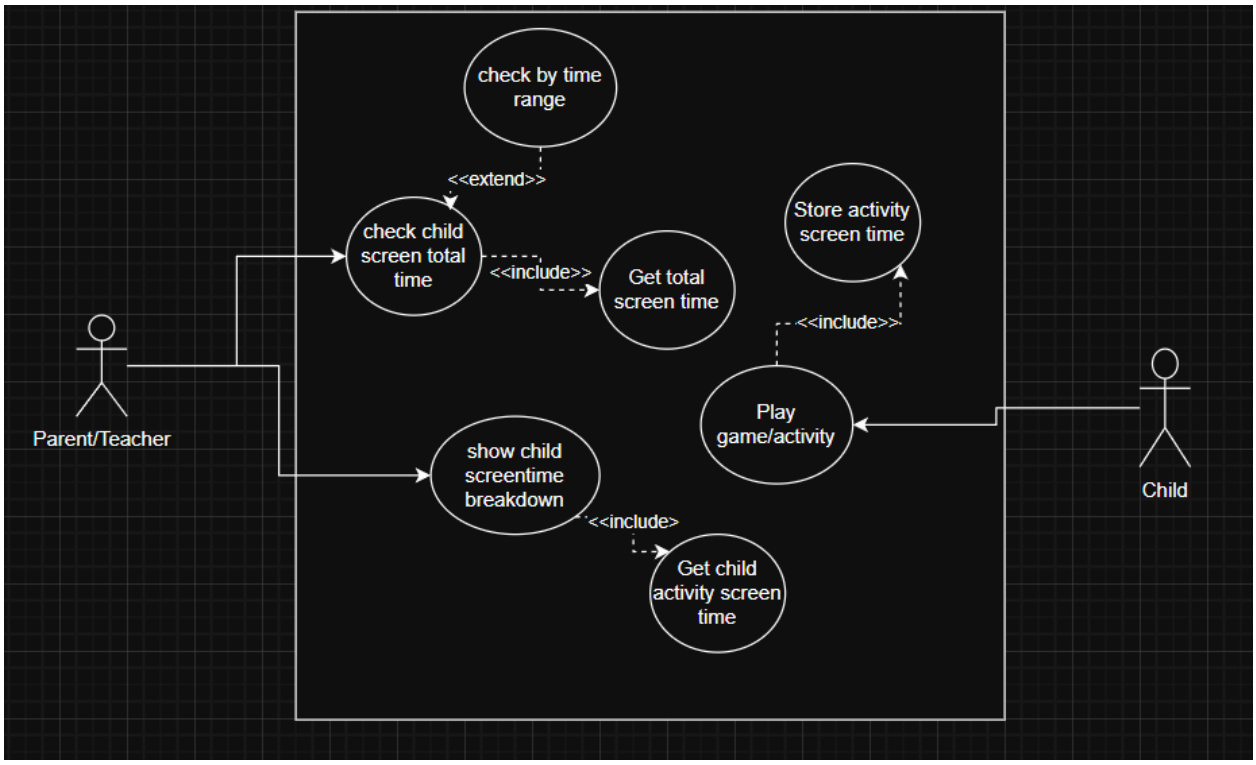
User Story
As a parent/teacher, I want to be able to check on the child to see how much time/ screen time he/she is spending on the activities, so that I can measure and manage how much time they are spending on those activities.

4.5.1) **Specifications**

ID	Specification
SP-5.1	The system must track and display the total screen time of the child.
SP-5.2	The system must give the option to display the total time spent in a specific time range ("... spent 3 hours in total this week")
SP-5.3	The system must track the screen time spent by the child in each game or activity
SP-5.4	The system must allow the parent/teacher to view a breakdown of time spent by the child in each game

	(this must be compatible with the time range feature)

4.5.2) **UC-05**



4.5.3) **Acceptance Criteria**

Acceptance Criterion	**Test Scenario**
The total screen time is calculated correctly.	Log in and play for exactly 20min, while using stopwatch, then check to see if it is correctly tracked.
All time range displays work correctly and accurately.	Log in and play for different time periods, while tracking time, and check after a month for example

	that the monthly, weekly and daily options work accurately.
System tracks time in each game accurately.	Play different games for different times and check if it is tracked and stored correctly.
The breakdown should show time spent in each game correctly.	Play different games for different times and periods, then check if it is displayed for each period correctly.

4.6) **FR6**: Allow parents/teachers to monitor a child's digital behavior

User Story

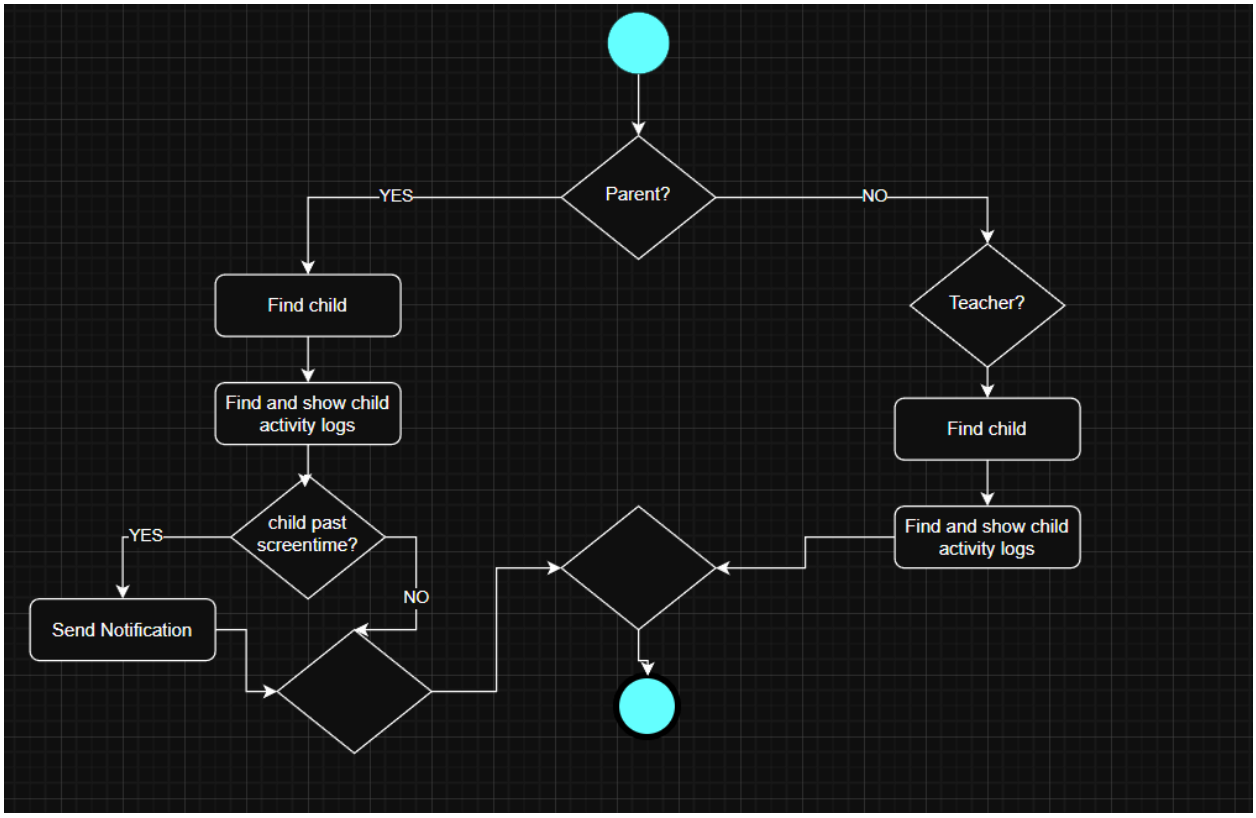
As a parent/teacher, I want to be able to check and monitor how the child is behaving and interacting with the platform, so that I can ensure and know if my child is learning and benefiting from the activities.

4.6.1) **Specifications**

ID	Specification
SP-6.1	The system must allow only parents ir teachers to monitor the child.
SP-6.2	The system must track which type of activities or subjects the child likes to play.

SP-6.3	The system must show teachers or parents to view logs of the child's activities
SP-6.4	The system must send a notification to the parent when the child has exceeded the screen time limit

4.6.2) **Activity Diagram**



4.6.3) **Acceptance Criteria**

Acceptance Criterion	Scenario
Children aren't able to use the monitoring feature.	Login as a child and check if able to access the monitoring feature.
The system stores logs of a child's activities correctly.	Play games as a child and then check if activities or behavior is stored correctly.
The parent/teacher is able to see the logs of the child correctly and accurately.	Login as parent/teacher and check if the logs appearing are correct, as in shown as stored and stored correctly.
System sends notification when child exceeds set screen time limits set by the parent.	Set screen time limit and exceed limit as child and check if notification is send to parent.

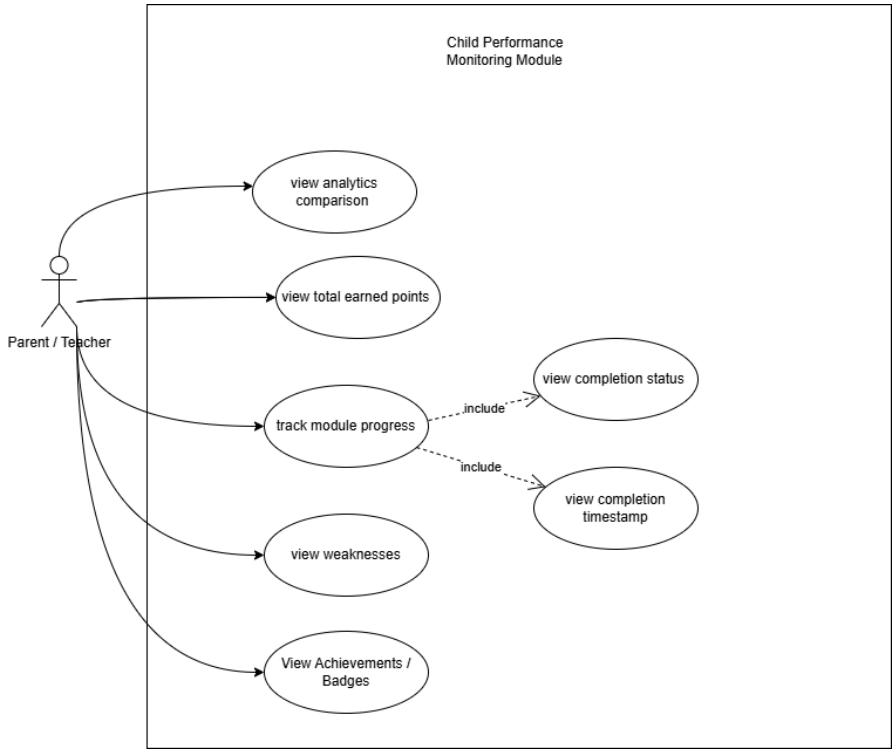
4.7) **FR-7:** Allow parents/teachers to monitor a child's performance.
User Story
As a parent/teacher, I want to monitor my child's performance so that I keep track of their progress and learning.

4.7.1) **Specifications**

ID	Specification
SP-7.1	The system should display analytics comparison between current and previous month.
SP-7.2	The system should display the total points earned by the child in the parent dashboard.
SP-7.3	The system should allow the parent to follow their child's progress in every module individually, displaying completion status and completion timestamp.

SP-7.4	The system should display weaknesses including delay in some modules and low total points for the child compared to average students points in the same age group in the dashboard of their parent.
SP-7.5	The system should allow parents to display the Achievements/badges of their child sorted descending where major achievements at the top.

4.7.2) **UC-07**



4.7.3) **Acceptance Criteria**

Acceptance Criterion	**Test Scenario**
Given a child has completed activities in multiple months, when the parent opens the dashboard, then the system shows a comparison between the current and previous month.	Complete modules in November and December → open parent dashboard → see side-by-side chart of November vs. December performance.
Given a child earns points, when the parent views the dashboard, then the total points are displayed correctly.	Child earns 50 points → parent opens dashboard → total points show updated score.
Given a child completes modules, when the parent checks progress, then each module's completion status and timestamp are visible.	Finish "Math Basics" module → parent checks dashboard → see module marked complete with timestamp.
Given a child has delays or low scores, when the parent views the dashboard, then delayed modules and below-average points are flagged.	Child completes only 2/5 modules → parent sees delayed modules highlighted and points compared to age group average.
Given a child earns achievements, when the parent opens the achievements section, then badges are listed with major achievements at the top.	Child unlocks "Math Master" and "Quick Learner" badges → parent opens achievements → "Math Master" appears first.

4.7.4) **WRSPM Diagram**



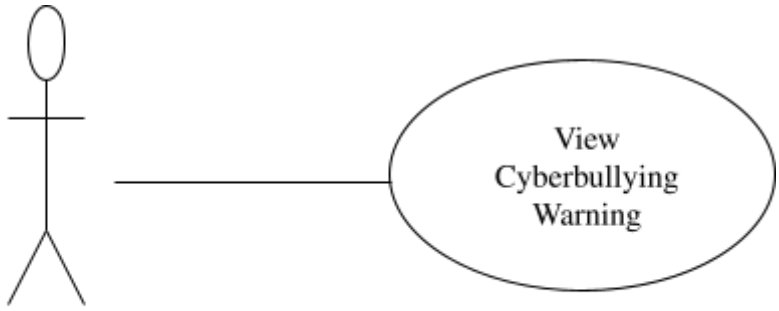
eh: Child's actual learning ability, parent's understanding of educational progress
ev: Child's point totals, module completion status, achievement badges earned
sv: Performance dashboard, comparison charts, warning indicators, achievement gallery
sh: Analytics algorithms, peer comparison calculations, achievement sorting logic

4.8) **FR-8**: Alert parents/teachers of potential cyberbullying.
User Story
As a parent/teacher, I want to get a warning if my child/student might face cyberbullying so I can act fast and keep them safe online.

4.8.1) **Specifications**

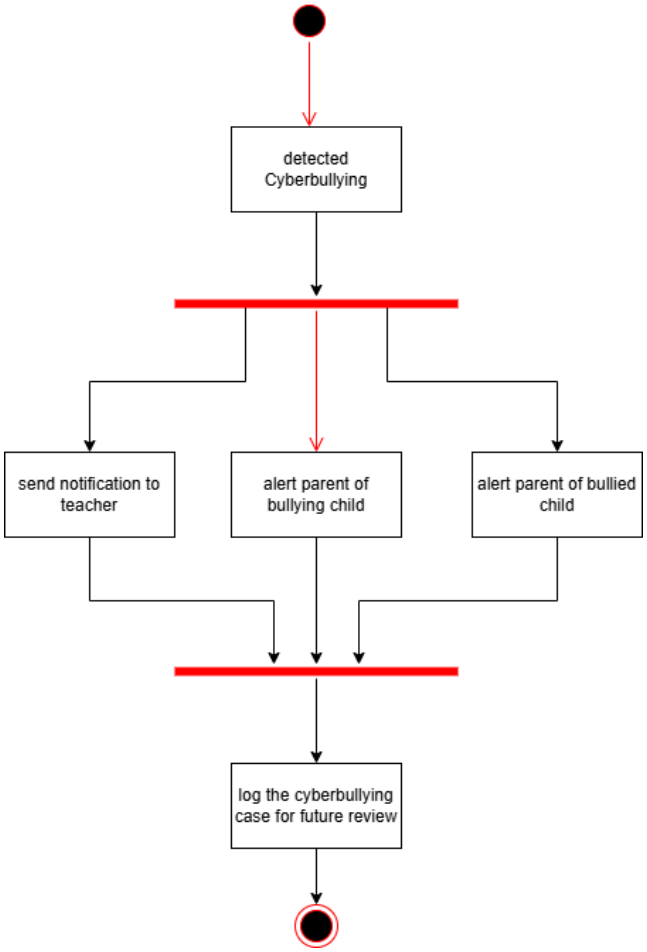
ID	Specification
SP-8.1	The system should send an immediate notification to the teachers after detecting a cyberbullying case.
SP-8.2	The system should send an immediate notification for the parents of both the child being bullied and the child who sent the bullying message .
SP-8.3	The notification should include time and context of the cyberbullying case .
SP-8.4	Cyberbullying history cases should be saved in a log so it can be retrieved and reviewed later

4.8.2) **UC-08**



Parent/teacher

4.8.3) **Activity diagram**



4.8.3) **Acceptance Criteria**

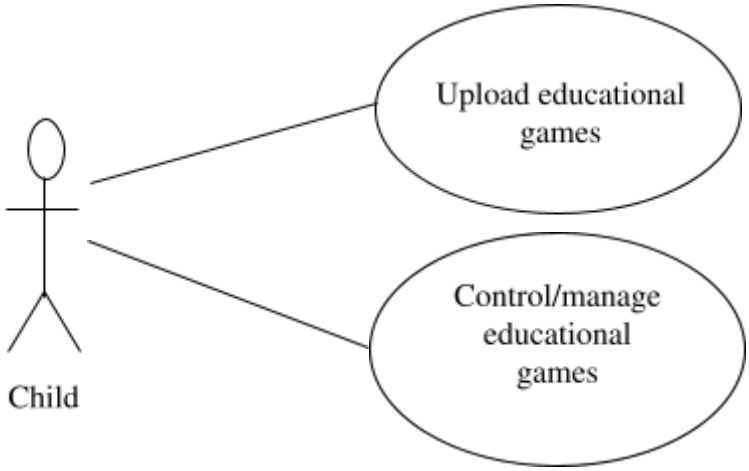
Acceptance Criterion	**Test Scenario**
Given a cyberbullying case, when the system identifies it, then it sends notifications immediately to the teacher.	Detect bullying message → system triggers → teacher receives notification immediately.
Given a cyberbullying case, when the system identifies it then a notification is sent to both parents of bullied child and bullying child.	Child A bullies Child B → system detects → parent of Child A and parent of Child B receive notification immediately.
Given a cyberbullying case identified by the system , then the notifications sent should include time and context of the case.	Bullying message sent at 10:15 am → notification generated → notification shows timestamp (10:15 am) and message context.
Given a cyberbullying case identified by the system , when the system logs events , then all cases are saved so they can be retrieved and reviewed later.	Child A bullied Child B → system logs case → later teacher opens logs → sees record with date, time, and context.

4.9) **FR-9:** Allow system administrators to upload and manage educational content.
User Story
As a system administrator, I want to upload and manage/control educational games . In other words, I want to have the ability to upload, edit, and delete educational games and learning materials in order for the system to be updated with new and relevant content geared towards children.

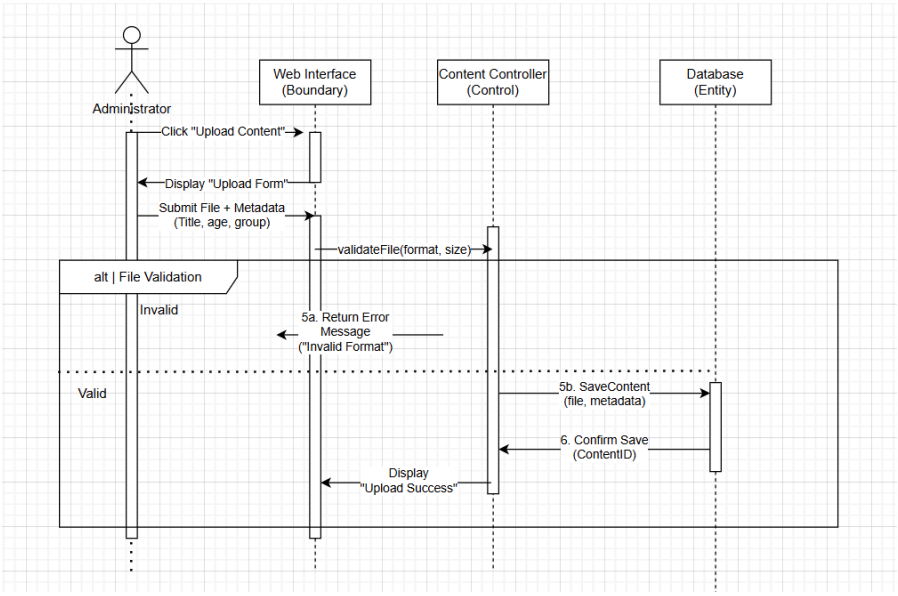
4.9.2) **Specification**

ID	**Specifications**
SP-9.1	In the platform, there should exist a dashboard only to a person with the “Administrator” role.
SP-9.2	The Administrator should have the ability to upload game assets with file size cap(e.g., 1GB) in certain formats (.zip, .html5).
SP-9.3	When uploading, the application will need the administrator to add metadata for every upload. For instance, he/she will need to input a description, title, targeted age group(3-5, 6-8, 9-12), and subject category (English, History, etc).
SP-9.4	While uploading, the application will check if the file format fits the desired/supported format. In other words, the file will get approved if it is in the supported format and it will be rejected if it has unsupported extensions.
SP-9.5	The administrator has the power to “Delete” or “Unpublish” content which will make it unavailable/unseen on the child’s dashboard.

4.9.3) **UC-09**



4.9.4) **Sequence Diagram**



4.9.5) **Acceptance Criteria**

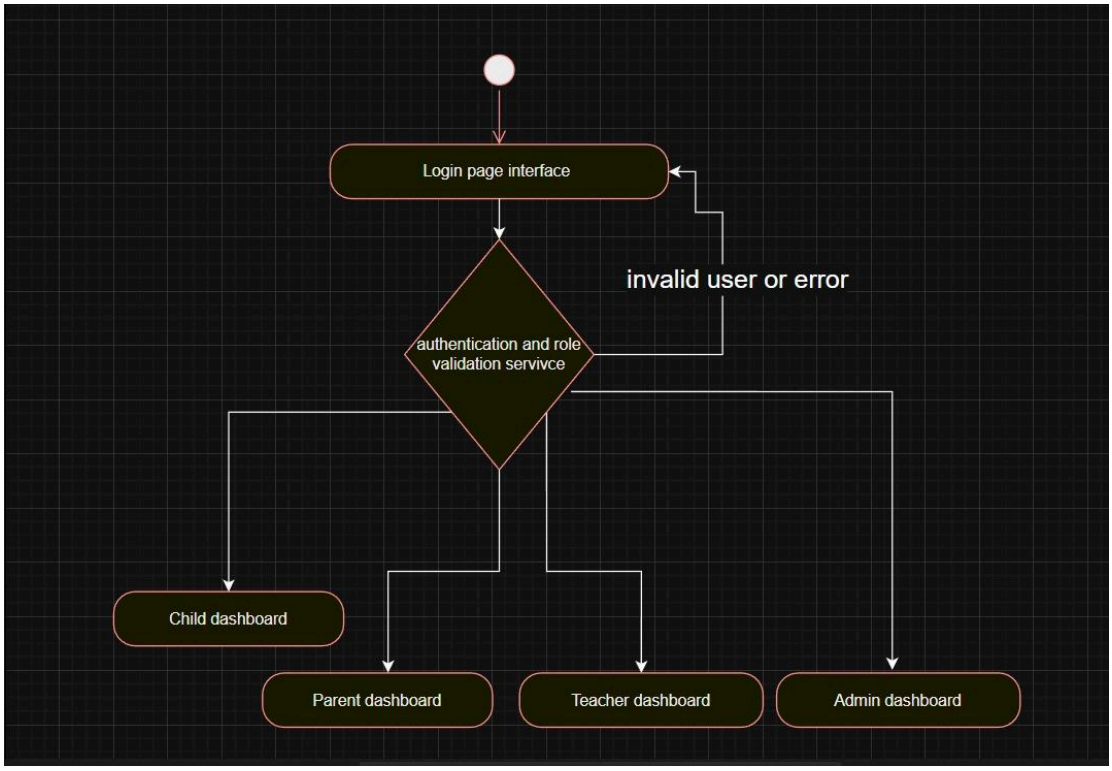
Acceptance Criterion	**Scenario**
Successful Content Upload	Since the administrator is already on the upload screen, when the uploaded file and metadata are all done, the system should then save the content and display “Upload Successful”.
Invalid File Type Rejection	After attempting to upload the file, the system denies the upload and displays an error “Invalid File Type”.
Content Removal	If the game is still working and the admin deletes the game, then the game should be removed from the database and it shouldn’t be displayed in the child’s search results.

4.10) **FR-10**: Multi-Role Login (Child, Parent, Teacher, Admin).
User story : as a the teacher submits the grades of the child ,also add the attendance of the child, and make a comment if there is a bad behaviour, as a child i should submit the quizzes and see the attendance if the teacher put it or not and in parent, in student i should see the grades of mine , as a admin i should see if there is bugs to fix it.

4.10.1) **Specification**

ID	Specification
SP-10.1	The structure for the system's login interface will be the same for every role. It will ask for the username/email and password. In the case of children within the age group of 3-5, it could be a 4-digit picture passcode.
SP-10.2	While logging in, the application shall query the database to get the user's defined role.
SP-10.3	For the routing logic, when a "Child" logs in, they are taken to the "Play & Learn" Game Lobby. Moreover, when a "Parent" logs in, they are taken to the "Parental Dashboard". When a "Teacher" logs in, they are taken to "Classroom Management". Lastly, when an Admin logs in, they are taken to "Content CMS".
SP-10.4	In order to prevent brute force attacks, the account will get locked for 15 minutes after 5 consecutive failed login attempts.
SP-10.5	For the convenience of preventing users from logging again after refreshing page, the timeout session will be 30 minutes.

4.10.2) **Activity Diagram**



4.10.3) **Acceptance Criteria**

Acceptance Criterion	**Scenario**
Role-Based Redirection	In the case a “Parent” is logging in, when they click “Login”, the browser URL then changes to /parent/dashboard and then the dashboard is visible
Role-Based Redirection	In the case a “teacher” is logging in, when they click “Login”, the browser URL then changes to /teacher/ClassRoomManagement and then the dashboard is visible
Role-Based Redirection	In the case a “admin” is logging in, when they click “Login”, the browser URL then changes to /admin/CMS and then the dashboard is visible
Child Simple Login	In the case a “Child” (Age 3-5) is logging in, when they select their avatar, they will then need to add their password to get to the game area.
Security Lockout	In the case a user enters their password incorrectly 5 times in a row, on the 6th try the system will tell them that the account is locked and they should try again in 15 minutes.

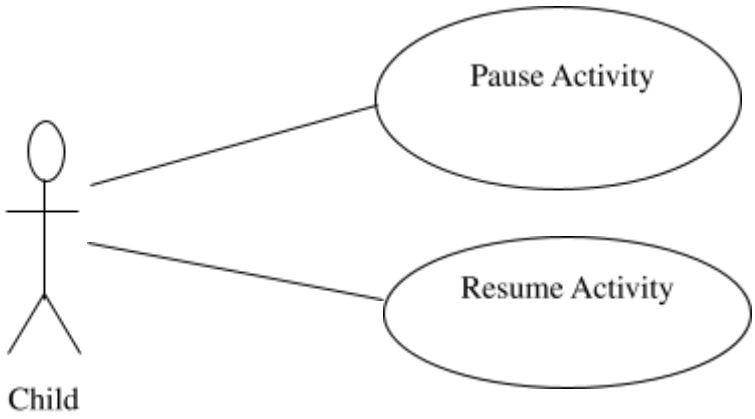
4.11) **FR-11**: Allow children to save and resume progress in learning modules.

As a child user, I want the game to remember exactly where I stopped, even if I close the tab or my internet disconnects. So that I can finish my food and come back later without having to start all over again.

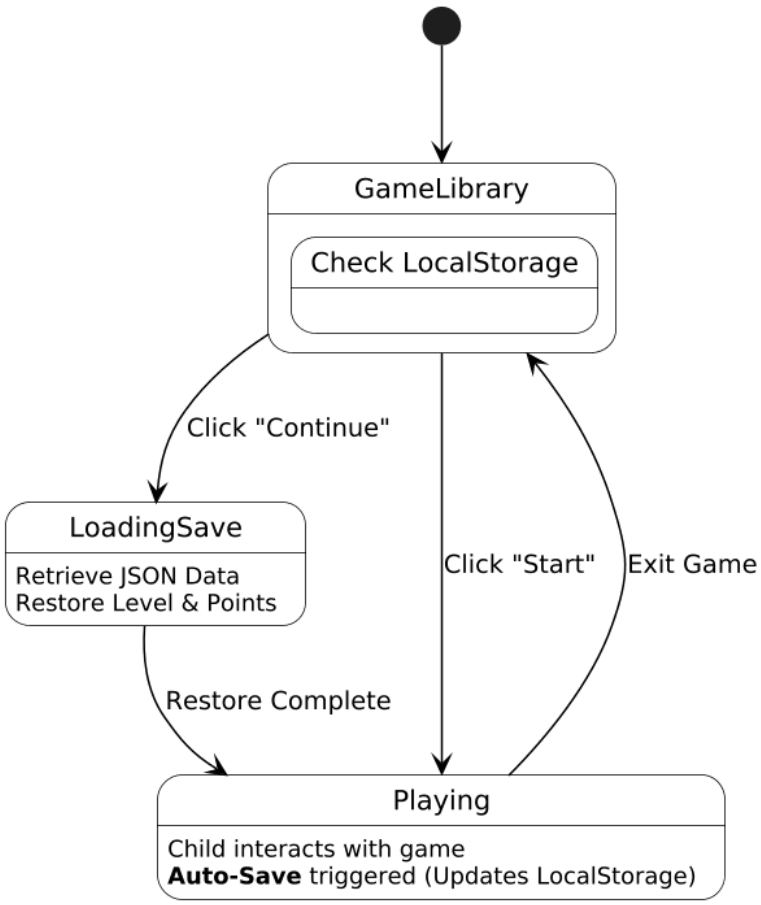
4.11.1) **Specifications**

ID	Specification	Notes
SP-11.1	The system must trigger an Auto-Save event at important moments. (ex. completing a level, earning a badge, or every 5 minutes during active gameplay.)	This ensures data isn't lost if the browser crashes or the parent enforces a "time's up" freeze.
SP-11.2	The "Save" object must include: current level ID, player position, current inventory, current points, and the timestamp.	For the prototype, data will be serialized to JSON and stored in the browser's Local Storage. This allows the "Resume" feature to work during the demo.
SP-11.3	A "Continue" button must appear on the game's cover card if a saved state exists.	If no save exists, the button should simply say "Play" or "Start".
SP-11.4	For the 3-5 age group, the system must visually show that a save is in progress while grabbing their attention (ex. a spinning star icon in the corner).	Young children need visual feedback to know it's safe to close the app.
SP-11.5	If a "Time Limit Reached" event occurs (ex. Parent settings), the system must force an immediate save before locking the screen.	Prevents punishment (losing progress) for following the rules.

4.11.2) **UC-11**



4.11.3) **State Diagram**



4.11.4) **Acceptance Criteria**

Acceptance Criterion	**Test Scenario**
Given that a child is in the middle of a level, if they close the tab suddenly, then their progress up to the last 5-minute checkpoint is restored.	Play for 7 minutes -> Close Tab -> Re-login -> Verify resume point is at minute 5.
Given that a child clicks the "Continue" button on a game, when the game loads, the player character is at the exact position and score they previously were on.	Move character to specific spot -> Logout -> Login/Continue -> Check character position.

Given that the daily screen time limit reaches, when the "Time's Up" screen appears, the backend checks that a final save state was recorded.	Set 1 min limit -> Wait for lock screen -> Check "save_timestamp".
Given that a child finishes a module, if they replay it later, they should have the option to "Retry" for a better score without affecting their Best Score.	Replay completed level -> Verify High Score remains visible on leaderboard.

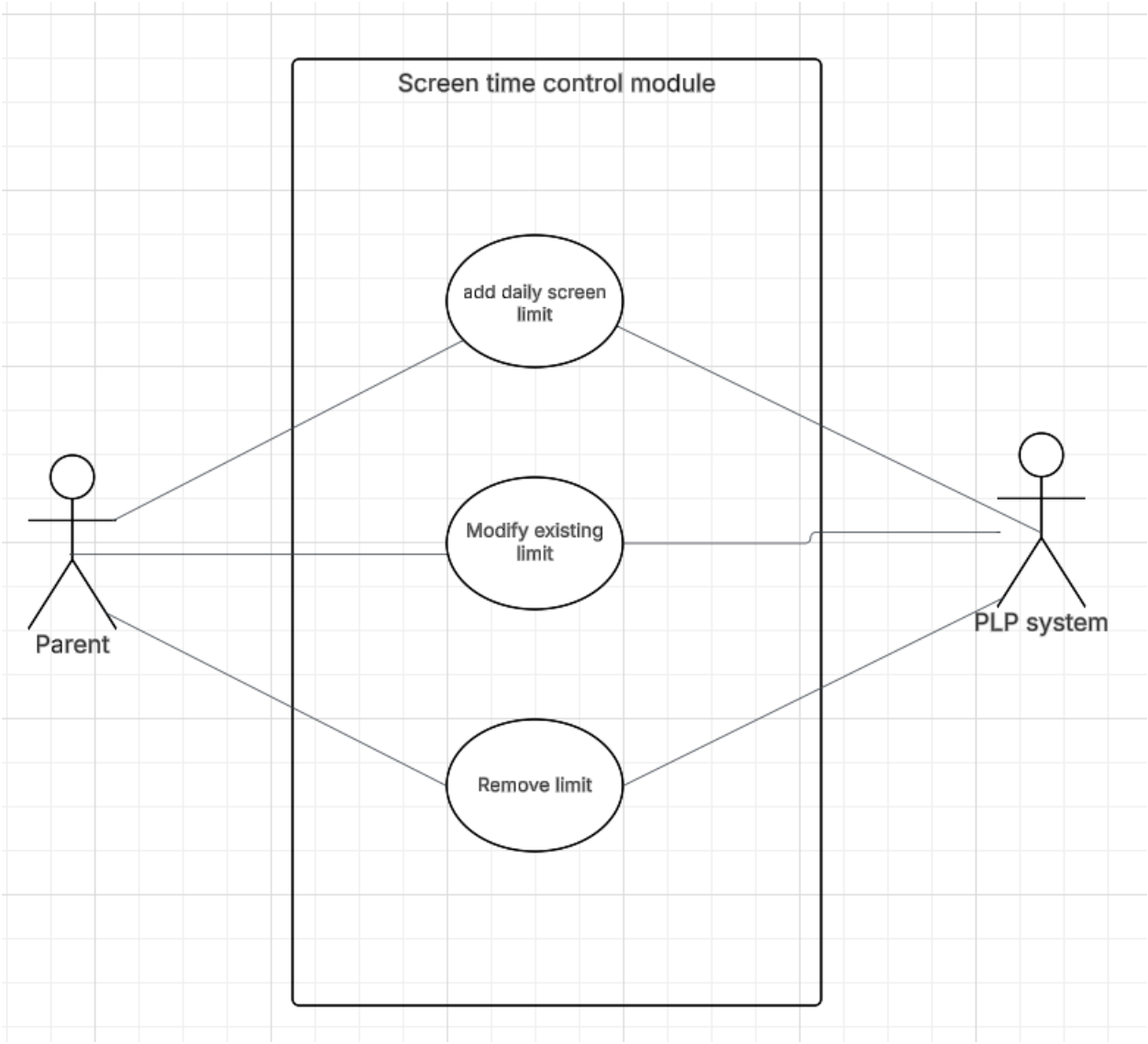
4.12) **FR-12**: Let parents set daily screen time limits per child account.
User story: As a parent, I want to be able to set a screen time limit for each child account, so that I can control their daily exposure on the app.

4.12.1) **Specifications**

ID	Specifications
SP-12.1	The system shall reset the screen time after every 24 hours have passed
SP-12.2	The system shall allows parent to modify daily screen time for any child at any time
SP-12.3	The system shall enforce the limits a parent set for the child once the daily limit is exceeded
SP-12.4	The system shall display a warning message to a child once the daily limit is reached

SP-12.5	The system shall notify the parent once a child has reached their daily limit

4.12.2) **UC-12**



4.12.3) **Acceptance Criteria**

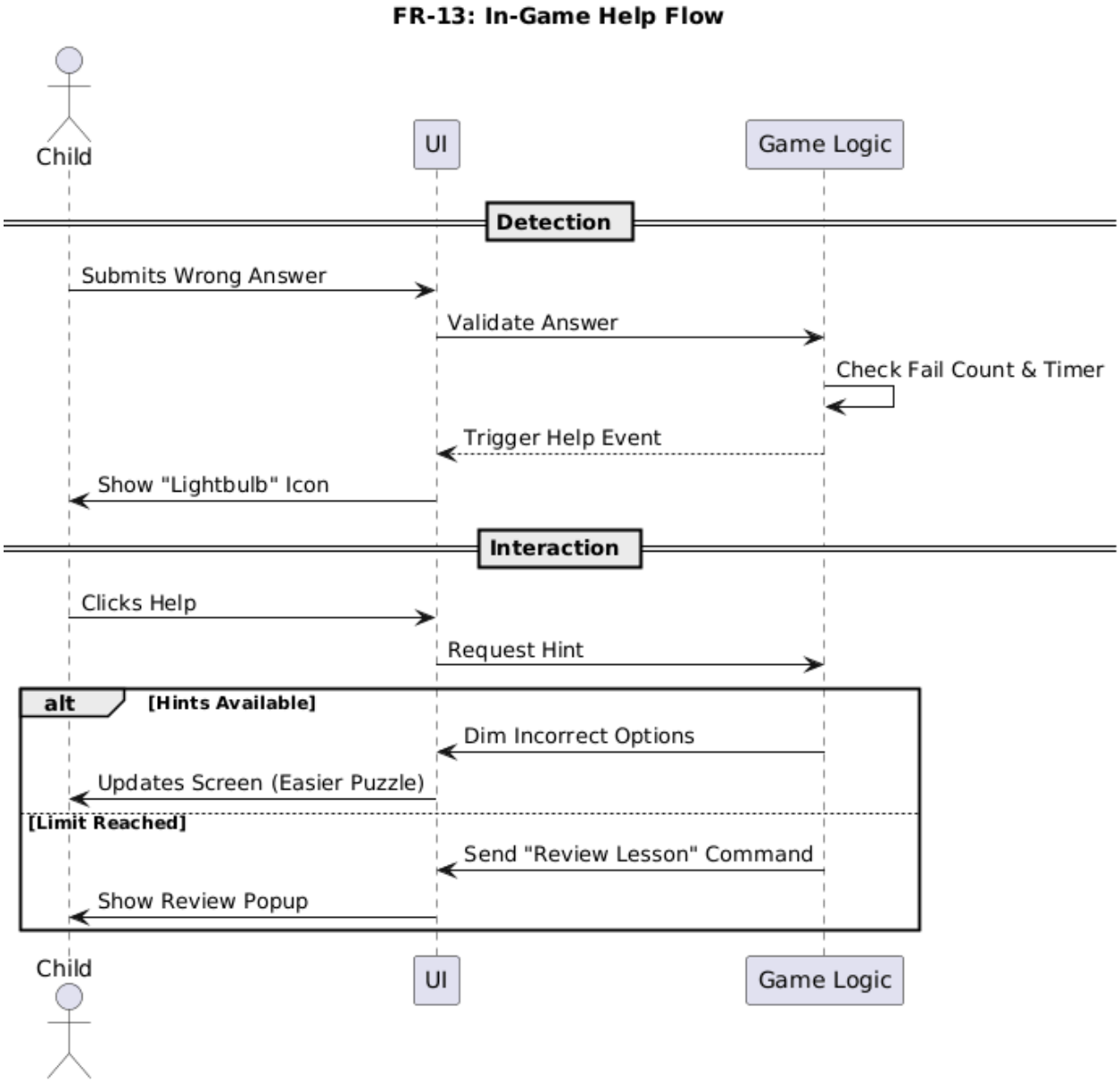
Acceptance Criterion	**Scenario**
Given that a parent sets a daily screen time limit for a child account, then the limit is saved and displayed correctly for that child.	The limit is set to 1 hour -> value is saved and displayed as 1 hour.
Given that a child exceeded their daily limit, subsequent access to the app is blocked.	Limit reached -> display message that limit is reached and block access.
Given that a new day starts, the screen time is reset to 0.	Use app at 5 p.m. today -> reset at 5 p.m. the next day.
Given that a child has a daily limit already set, a parent can update it and the new limit takes effect immediately.	Parent reduces limit below already used time -> child is blocked immediately.

4.13) **FR-13**: Provide help within games when a child is stuck.
User Story: As a child user (especially ages 3-5), I want a friendly character to give me a hint if I stop moving or keep getting the answer wrong, so that I don't feel sad or get frustrated and quit the game.

4.13.1) **Specifications**

ID	Specification	Notes
SP-13.1	The system must detect "Stuck Behavior". (ex. 3 incorrect attempts at a puzzle,45 seconds of total inactivity.)	Time should be lower for the 3-5 age group (ex. 20 seconds).
SP-13.2	Upon detecting "Stuck Behavior," a visual "Help" indicator (ex. a glowing lightbulb) should appear.	It shouldn't interrupt gameplay forcefully, just offer assistance.
SP-13.3	When the help feature is chosen, the system should reduce the puzzle's difficulty by removing or dimming incorrect answers.	This approach helps the child focus without giving away the answer, which is excellent for the Play & Learn educational goals.
SP-13.4	Using the "Help" feature should be recorded in the Child's Performance Report.	Teachers need to see if a student relies alot on hints for specific topics for extra assistance.
SP-13.5	To prevent spamming hints to win, the system will limit hints to 3 per game level.	If 3 hints are used, the system suggests a "Review Lesson" instead.

4.13.2) **Sequence Diagram**



4.13.3) **Acceptance Criteria**

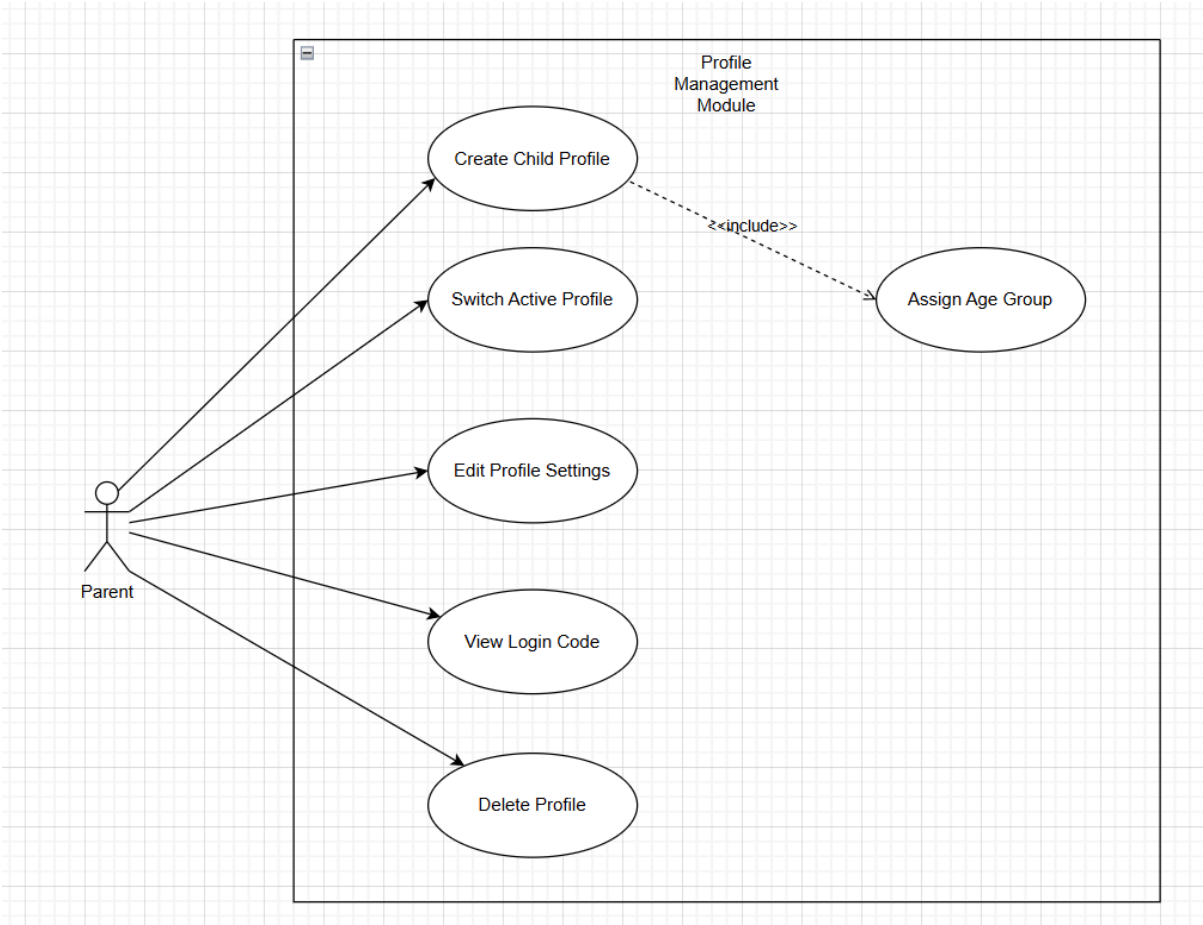
Acceptance Criterion	**Test Scenario**
Given that a child answers a math question 3 times wrong,the helper mascot appears with a suggestion.	Enter wrong answer 3 times -> check tht mascot animation happens.
Given that a child uses a hint during an interactive game, when the hint happens, the correct element on the screen visually appears to grab their attention.	Start a 'Match the Shapes' game -> Click the 'Help' button -> Check that the correct matching shape starts glowing.
Given that a child uses a hint, when the level is finished, the Teacher Dashboard shows that a hint was used for that problem.	Use hint -> Complete level -> Log in as Teacher -> Check Student Report.
Given that the child clicks the help button alot of times, when the limit (3) is reached, the button is disabled.	Click help 4 times -> Check 4th click does nothing or shows "No more hints".

4.14) **FR-14**: Allow parental accounts to manage multiple child profiles
User Story
As a Parent, I want to be able to add several child profiles to my account so I can control their settings and see the progress for all of my children from one login.

4.14.1) **Specifications**

ID	Specification
SP-14.1	The parent should have the ability to create their child's profile by clicking "Add Child" from their(parent's) dashboard.
SP-14.2	In order for the parent to create their child's profile, the system will require for the parents to add their name, age, and avatar preference.
SP-14.3	By analyzing the child's information from the parent's inputs, the application will assign a child to their age group.
SP-14.4	The parent will be able to switch between their children' s profiles to view each child's reports without logging out.
SP-14.5	Each child profile will have a simple username or a unique login code which is generated by the application.

4.14.2) **UC-14**



4.14.3) **Acceptance Criteria**

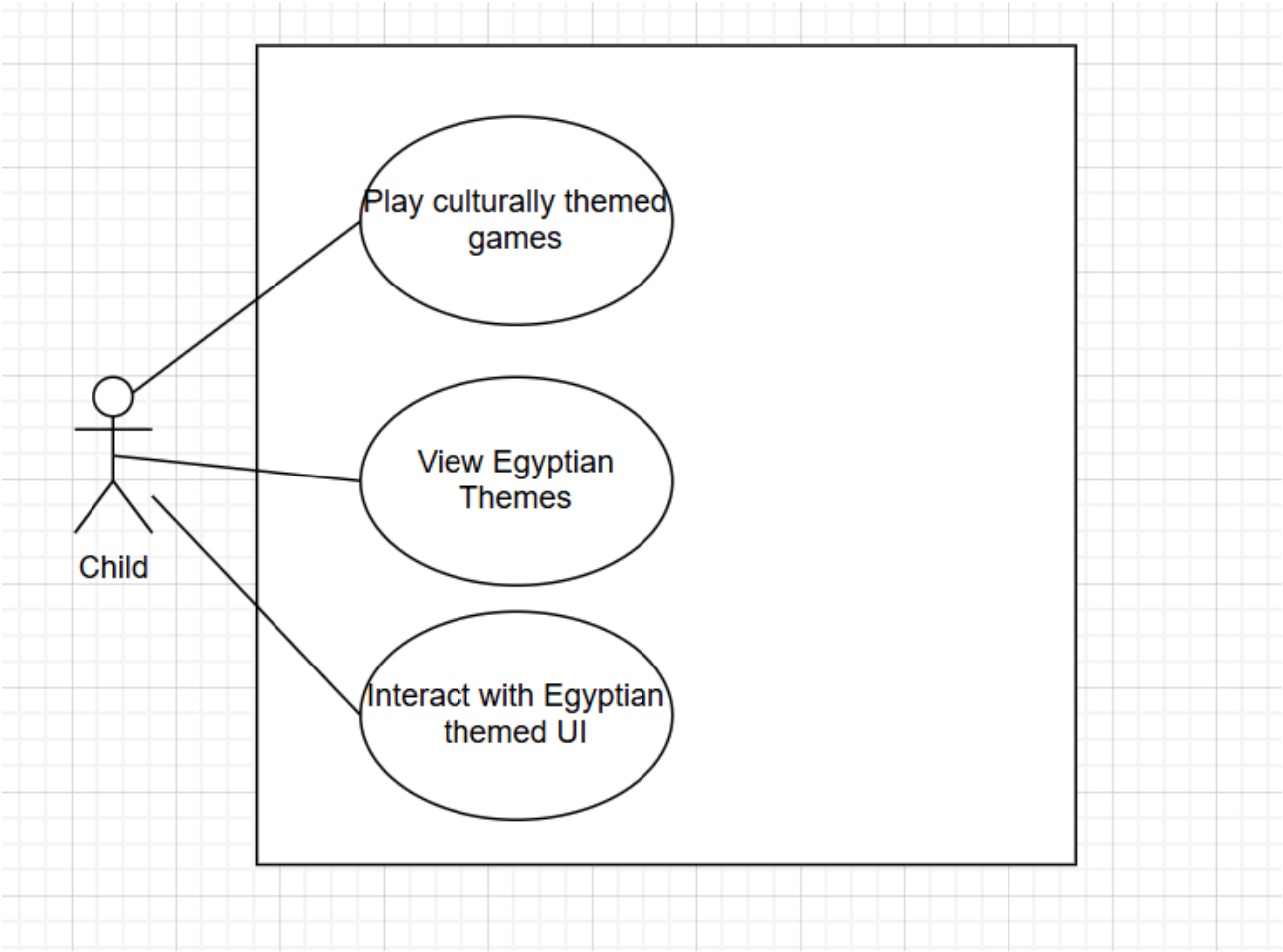
Acceptance Criterion	**Scenario**
Child Login Redirect	If a child user logs into the application, he/she will be taken to the Game Library page.
Parent Login Redirect	If a parent user logs into the application, he/she will be taken to the parent dashboard.
Invalid Login	If a user enters the wrong password when they are logging in, the application will stay on the login page and it will display “Invalid Username or Password.”

4.15) FR-15 Include culturally relevant themes games
User Story: **As a** child in Egypt,
I want to play games that reflect my own culture, history, and environment,
so that I can learn and have fun while seeing my own world represented and valued.

4.15.1) **Specifications**

ID	**Specification**
SP-15.1	Game themes, characters, and settings shall be based on Egyptian culture
SP-15.2	Character design shall reflect Egyptian diversity in appearance, names, and attire, ensuring positive representation.
SP-15.3	Audio shall include traditional Egyptian music and instruments
SP-15.4	Visuals shall use color palettes and artistic styles inspired by Egyptian art

4.15.2) **UC-15**



4.15.3) **Acceptance Criteria**

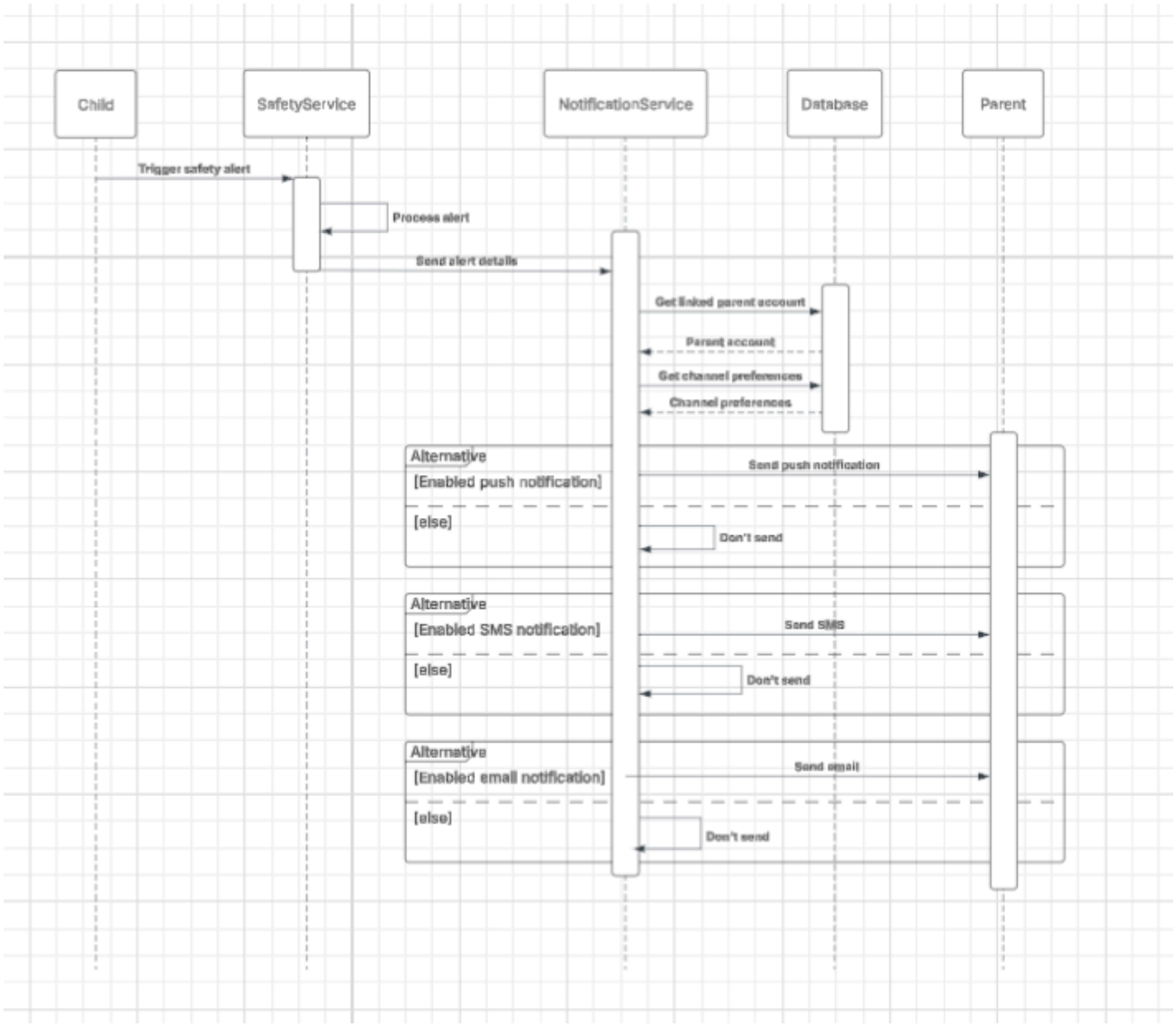
Criteria	**Test Case**
Given a child opens the game library, when they browse games, then at least 90% of displayed games feature Egyptian cultural themes.	Open library → count Egyptian-themed games vs. total → verify ≥90%.
Given a game loads, when the child sees characters, then character names, clothing, and appearances reflect Egyptian diversity.	Launch game → inspect characters → verify names (e.g., Ahmed, Fatima) and attire are culturally appropriate.
Given a game includes audio, when played, then traditional Egyptian music or instruments are present in the soundtrack.	Play game → listen to audio → identify at least one traditional instrument (e.g., oud, ney).

4.16) **FR-16**: Notify parents when a safety alert is triggered.
User story: As a parent I want to be notified when a safety alert is triggered, so that I can take immediate action to protect my child.

4.16.1) **Specification**

ID	Specification
SP-16.1	The system shall deliver notifications in real time after the alert is triggered.
SP-16.2	The system shall ensure only linked parent account is notified
SP-16.3	The system shall notify through multiple channels (e.g., push notification, SMS, email)
SP-16.4	The system shall include important details in the alert notification (alert type, child account, timestamp)

4.16.2) **Sequence diagram**



4.16.3) **Acceptance Criteria**

Acceptance Criterion	Scenario
Given that a child triggers a safety alert, a notification is sent to the linked parent account	Trigger a safety alert → parent receives notification
Given that a safety alert is triggered, the alert must be processed and notifies the parent within a defined threshold	Trigger a safety alert -> alert processed and notification sent within 5 seconds
Given that a parent has configured notification channel preferences, notifications are only sent through the enabled channels	Push notification enabled, but SMS disabled -> only push notification is sent
Given that a notification fails to deliver, the system detects and corrects by retrying	Network failure -> system retries

5. Nonfunctional Requirements

5.1 Performance Requirements

In order to provide an engaging experience, the platform must provide responsive and fluid interactions across all user pages. Below are requirements that ensure the system performs fluidly and responsibly under various scenarios.

ID	Description
NFR-01	The system must be able to handle at least 1,000 concurrent users without any noticeable delays
NFR-02	Under normal conditions, all webpages and dashboards must load in under three seconds
NFR-03	API responses must be under 100 milliseconds
NFR-04	Realtime updates must propagate to clients under 1 second

5.2 Safety Requirements

The system must protect children from digital harm. Below are requirements that ensure users are protected and secure.

ID	Description
NFR-05	All user-generated content, chat, and comments must be processed in real time for Cyberbullying, hate speech, and inappropriate material using Google's Perspective APIs
NFR-06	Parents must be able to set daily time limits for their child's account
NFR-07	Upon reaching a limit, the platform must automatically pause the session
NFR-08	Data regarding users name, age, location must not be visible to anyone without parental approval
NFR-09	An "Account Freeze" function must be available to parents to immediately suspend their child's access

5.3 Security Requirements

ID	Description
NFR-10	Passwords should have a number of conditions that are set in place
NFR-11	Multi-factor authentication for parent and teacher accounts
NFR-12	Child accounts can only be created through a parent
NFR-13	Teacher accounts require verification via a valid institutional email and approval by the administrator
NFR-14	Sensitive data at must be encrypted using AES-256
NFR-15	Use a role-based access control with the following roles: Child, Parent, Teacher, Administrator

5.4 Usability

ID	Description
NFR-16	A child must be able to navigate to and launch a game within 3 interactions, without external help.
NFR-17	A parent or teacher with basic digital literacy must be able to navigate the application with ease without requiring user manuals or support.

5.5 Reliability & Availability:

ID	Description
NFR-18	The platform must maintain 99% uptime during core operational hours (6:00 AM - 10:00 PM local time).
NFR-19	Scheduled maintenance must occur outside of peak child usage hours (e.g., 2:00 AM - 6:00 AM)
NFR-20	System failures must fail gracefully